

Data Science with Python Internship – Task 3

Task Description

In Task-2, you analyzed the **Titanic dataset** for survival by gender and class.

Now, let's perform a **Mini Exploratory Data Analysis (EDA)** on the Titanic dataset with **more advanced visualizations**.

You will:

- Handle missing values better (imputation).
- Use groupby to find deeper insights.
- Create multiple visualizations to tell a story.

What You'll Build (Requirements)

1. Clean dataset:
 - Fill missing Age with mean.
 - Drop irrelevant columns (like Cabin).
2. Analysis Questions:
 - Survival rate by **Age Group**.
 - Survival rate by **Embarkation Port**.
 - Survival rate by **Family size (SibSp + Parch)**.
3. Visualizations:
 - Age distribution (histogram).
 - Heatmap of correlations.
 - Survival by family size (bar plot).

Deliverable

A **Jupyter Notebook** with cleaned data, multiple insights, and professional visualizations.

Why This Task Is Important

- Teaches you **EDA (Exploratory Data Analysis)**, a must for every data scientist.
- Improves your **data cleaning skills**.
- Helps you build **storytelling with data**.

Sample Starter Code

python

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# Load dataset
df = pd.read_csv("titanic.csv")

# Fill missing Age values with mean
df["Age"].fillna(df["Age"].mean(), inplace=True)

# Drop irrelevant columns
df.drop(columns=["Cabin"], inplace=True)

# Create Age Groups
df["AgeGroup"] = pd.cut(
    df["Age"],
    bins=[0, 12, 18, 30, 50, 80],
    labels=["Child", "Teen", "YoungAdult", "Adult", "Senior"]
)

# Survival by Age Group
sns.barplot(x="AgeGroup", y="Survived", data=df)
plt.title("Survival Rate by Age Group")
plt.show()

# Heatmap of correlations
sns.heatmap(df.corr(), annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()

# Create Family Size feature
df["FamilySize"] = df["SibSp"] + df["Parch"]

# Survival by Family Size
sns.barplot(x="FamilySize", y="Survived", data=df)
plt.title("Survival Rate by Family Size")
plt.show()
```

Free Video Courses / Tutorials

- [Data Science Crash Course – FreeCodeCamp](#)
- [Pandas Crash Course – Corey Schafer](#)
- [Seaborn Visualization Crash Course](#)

Documentation

- [Pandas Docs](#)
- [Matplotlib Docs](#)
- [Seaborn Docs](#)

Tools

- [Google Colab \(Free Jupyter Notebook\)](#)
- [Kaggle Datasets](#)
- [Titanic Dataset](#)